

1. What would happen if you skipped the differential backup step?

If you skip the **differential backup** step in a restore chain, you will **lose all changes** that occurred after the last **full backup** and **before** the most recent **transaction log backups**.

Explanation:

- A differential backup contains all changes since the last full backup.
 - Transaction log backups record changes since the last log backup, but they require the most recent differential or full backup to make sense of the transaction sequence.
 - Skipping the differential step can break the restore chain and prevent a successful point-in-time recovery unless you only rely on the full + all logs (which is less efficient).
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2. What's the difference between restoring a full vs. copy-only backup?

A **full backup** becomes the **base** for future **differential and log backups**, while a **copy-only backup** is an **independent backup** that does **not affect the backup chain**.

Key differences:

- **Full Backup:**
 - Updates the database's backup history.
 - Impacts future differential backups.
 - **Copy-Only Backup:**
 - Does *not* update the backup history.
 - Useful for ad-hoc backups (e.g., before a risky operation) without affecting the regular backup cycle.
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3. What happens if you use WITH RECOVERY in the middle of a restore chain?

Using **WITH RECOVERY** in the middle of a restore chain will **end the restore process prematurely** and **roll forward the database**, making it **unusable for applying additional backups**.

Explanation:

- WITH RECOVERY makes the database **available for use** by undoing uncommitted transactions.
 - This action **closes the restore sequence**, so no further differential or log backups can be applied.
 - To continue a restore chain, intermediate steps must use **WITH NORECOVERY**, reserving **WITH RECOVERY** only for the final restore.
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4. Which backup types are optional and which are mandatory for full recovery?

Mandatory for Full Recovery:

- **Full backup** – Required as the baseline.
- **Transaction log backups** – Needed for point-in-time recovery.

Optional but Recommended:

- **Differential backup** – Optional but useful to speed up restores by reducing the number of log backups needed.
- **Copy-only backup** – Optional, mainly for special scenarios.